

Genome Sequencing

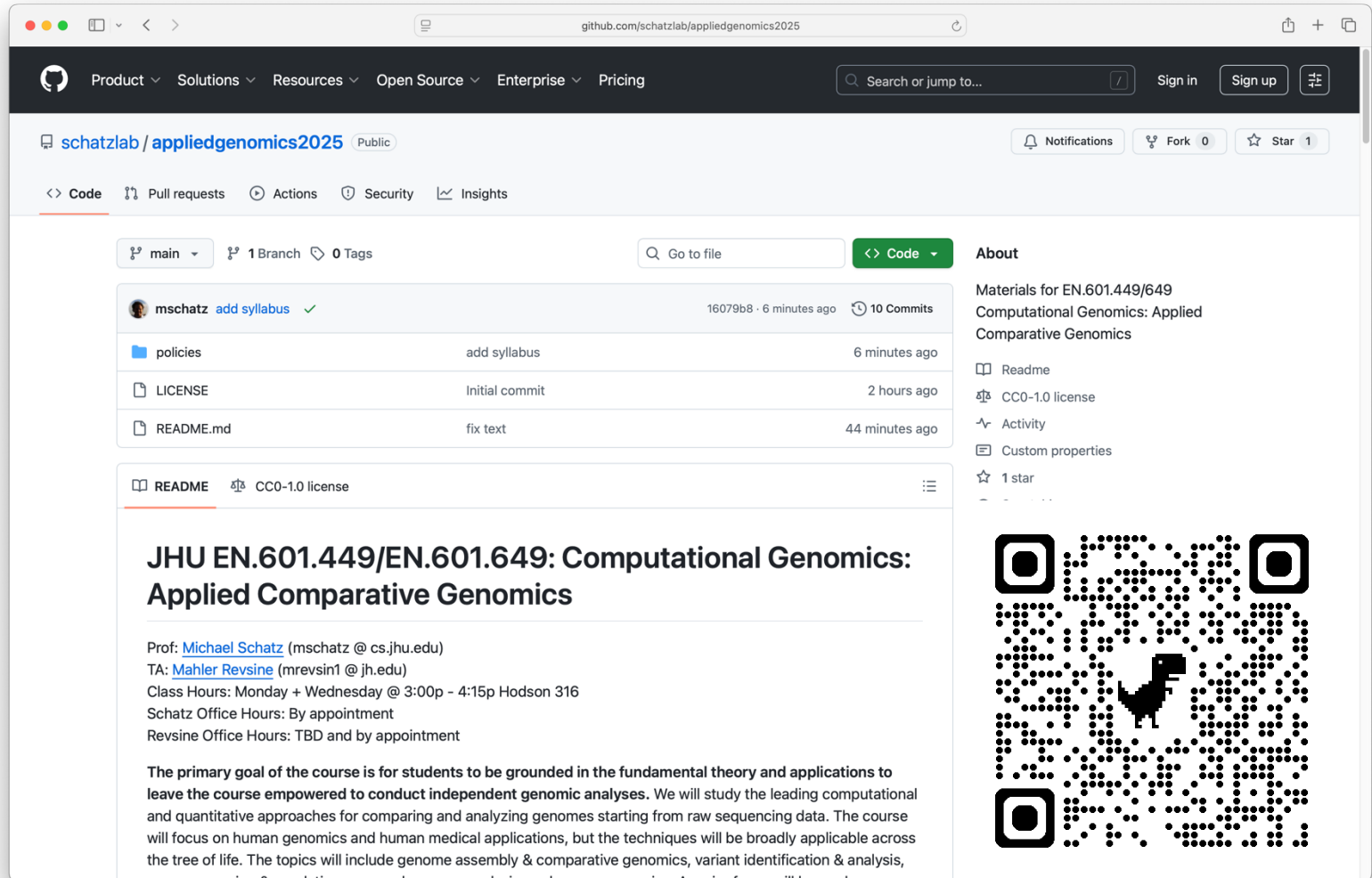
Michael Schatz

August 27 2025

Lecture 2: Applied Comparative Genomics



Course Webpage



github.com/schatzlab/appliedgenomics2025

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mschatz add syllabus 16079b8 · 6 minutes ago 10 Commits

polices	add syllabus	6 minutes ago
LICENSE	Initial commit	2 hours ago
README.md	fix text	44 minutes ago

README CC0-1.0 license

JHU EN.601.449/EN.601.649: Computational Genomics: Applied Comparative Genomics

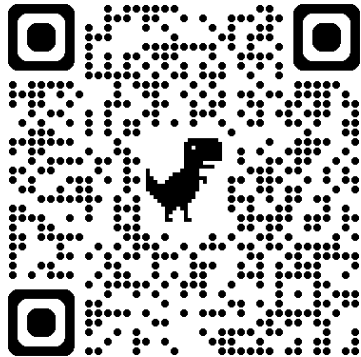
Prof: [Michael Schatz](mailto:mschatz@cs.jhu.edu) (mschatz@cs.jhu.edu)
TA: [Mahler Revsine](mailto:mrevsin1@jh.edu) (mrevsin1@jh.edu)
Class Hours: Monday + Wednesday @ 3:00p - 4:15p Hodson 316
Schatz Office Hours: By appointment
Revsine Office Hours: TBD and by appointment

The primary goal of the course is for students to be grounded in the fundamental theory and applications to leave the course empowered to conduct independent genomic analyses. We will study the leading computational and quantitative approaches for comparing and analyzing genomes starting from raw sequencing data. The course will focus on human genomics and human medical applications, but the techniques will be broadly applicable across the tree of life. The topics will include genome assembly & comparative genomics, variant identification & analysis, gene expression & regulation, personal genome analysis, and cancer genomics. A major focus will be on deep

About

Materials for EN.601.449/649
Computational Genomics: Applied
Comparative Genomics

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Custom properties
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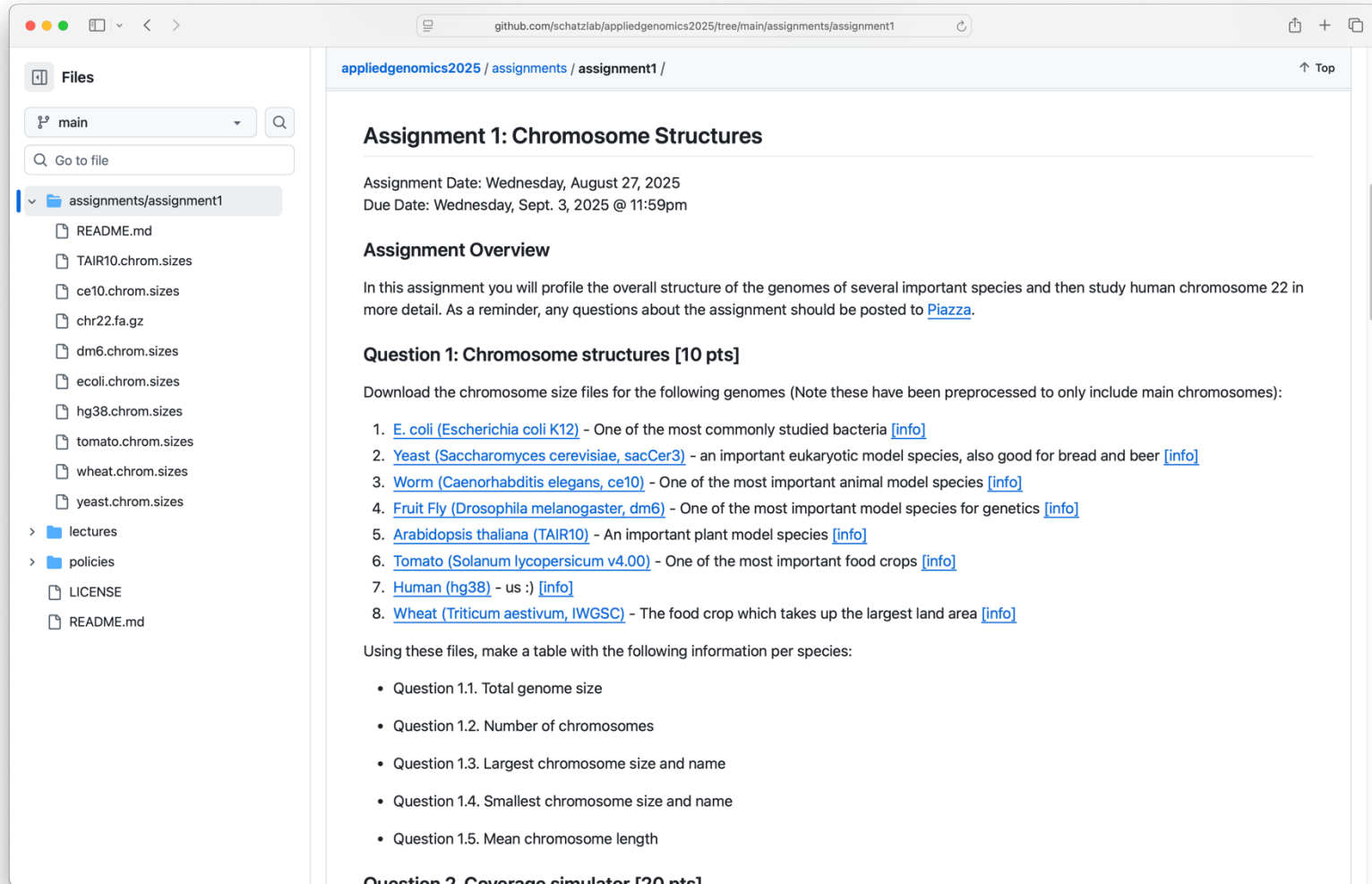
<https://github.com/schatzlab/appliedgenomics2025>

TA: Mahler Revsine



Office Hours: Mondays at 4:30pm
@ Malone 216 and Zoom

Assignment I



The screenshot shows a web browser displaying a GitHub repository page. The address bar shows the URL: `github.com/schatzlab/appliedgenomics2025/tree/main/assignments/assignment1`. The page title is `appliedgenomics2025 / assignments / assignment1`. On the left, a 'Files' sidebar shows a directory tree with 'main' selected. Under 'assignments/assignment1', there is a list of files: `README.md`, `TAIR10.chrom.sizes`, `ce10.chrom.sizes`, `chr22.fa.gz`, `dm6.chrom.sizes`, `ecoli.chrom.sizes`, `hg38.chrom.sizes`, `tomato.chrom.sizes`, `wheat.chrom.sizes`, and `yeast.chrom.sizes`. Below these are folders for 'lectures' and 'policies', and files for 'LICENSE' and 'README.md'. The main content area has the heading 'Assignment 1: Chromosome Structures'. It includes the assignment date (Wednesday, August 27, 2025) and due date (Wednesday, Sept. 3, 2025 @ 11:59pm). An 'Assignment Overview' section states the goal is to profile genome structures and study human chromosome 22. A 'Question 1: Chromosome structures [10 pts]' section instructs to download chromosome size files for eight species, providing links to their genome data. A list of questions follows, asking for total genome size, number of chromosomes, largest and smallest chromosome size and name, and mean chromosome length for each species. The start of 'Question 2: Coverage simulator [20 pts]' is visible at the bottom.

Assignment 1: Chromosome Structures

Assignment Date: Wednesday, August 27, 2025
Due Date: Wednesday, Sept. 3, 2025 @ 11:59pm

Assignment Overview

In this assignment you will profile the overall structure of the genomes of several important species and then study human chromosome 22 in more detail. As a reminder, any questions about the assignment should be posted to [Piazza](#).

Question 1: Chromosome structures [10 pts]

Download the chromosome size files for the following genomes (Note these have been preprocessed to only include main chromosomes):

1. [E. coli \(Escherichia coli K12\)](#) - One of the most commonly studied bacteria [\[info\]](#)
2. [Yeast \(Saccharomyces cerevisiae, sacCer3\)](#) - an important eukaryotic model species, also good for bread and beer [\[info\]](#)
3. [Worm \(Caenorhabditis elegans, ce10\)](#) - One of the most important animal model species [\[info\]](#)
4. [Fruit Fly \(Drosophila melanogaster, dm6\)](#) - One of the most important model species for genetics [\[info\]](#)
5. [Arabidopsis thaliana \(TAIR10\)](#) - An important plant model species [\[info\]](#)
6. [Tomato \(Solanum lycopersicum v4.00\)](#) - One of the most important food crops [\[info\]](#)
7. [Human \(hg38\)](#) - us :) [\[info\]](#)
8. [Wheat \(Triticum aestivum, IWGSC\)](#) - The food crop which takes up the largest land area [\[info\]](#)

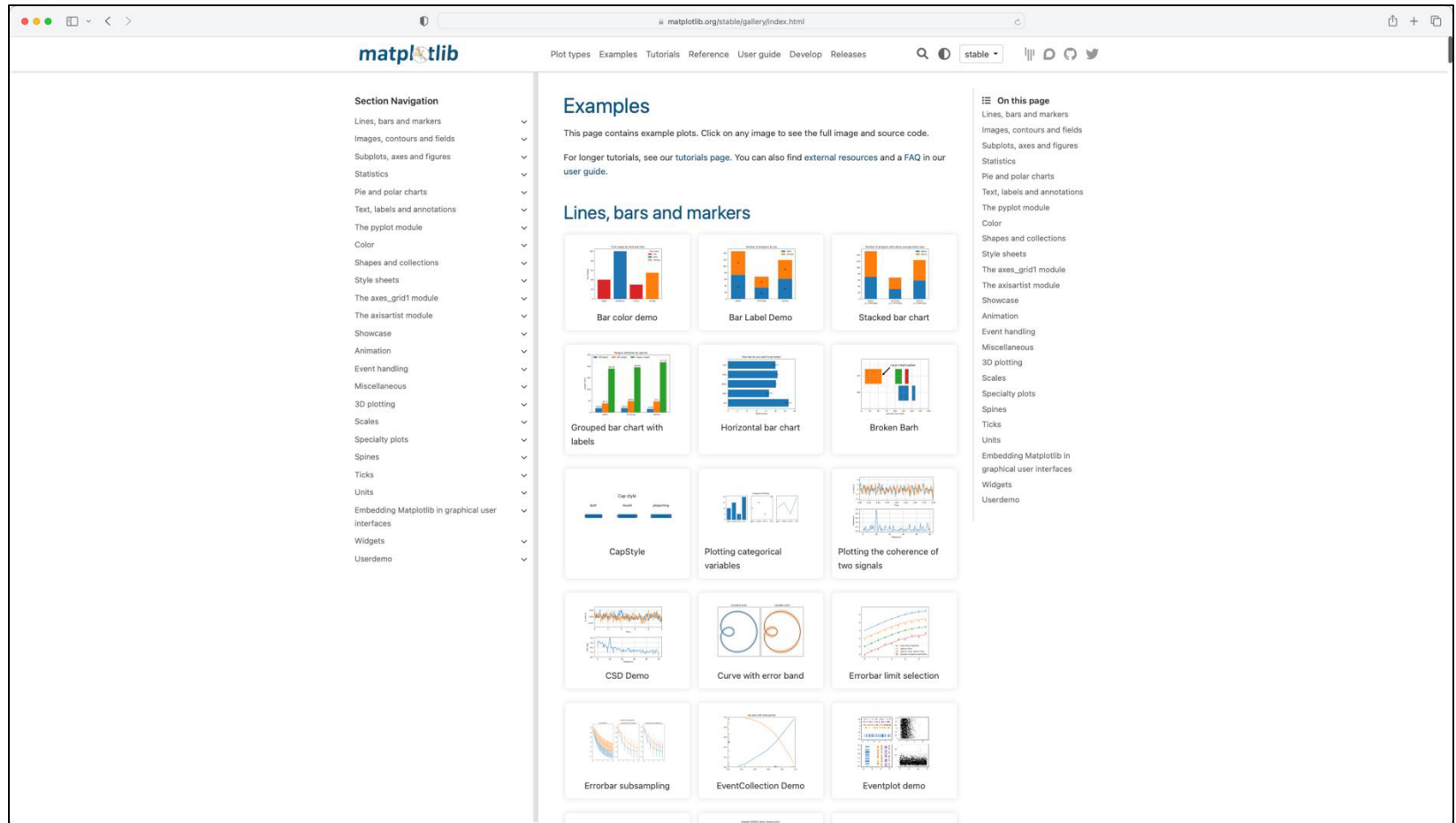
Using these files, make a table with the following information per species:

- Question 1.1. Total genome size
- Question 1.2. Number of chromosomes
- Question 1.3. Largest chromosome size and name
- Question 1.4. Smallest chromosome size and name
- Question 1.5. Mean chromosome length

Question 2: Coverage simulator [20 pts]

<https://github.com/schatzlab/appliedgenomics2025/tree/main/assignments/assignment1>
Due end of day on Wednesday Sept 3 (right before midnight)

Plotting in Python



<https://matplotlib.org/>

Plotting in R / ggplot2

Data visualization with ggplot2 : : CHEATSHEET



Basics

ggplot2 is based on the **grammar of graphics**, the idea that you can build every graph from the same components: a **data set**, a **coordinate system**, and **geoms**—visual marks that represent data points.



To display values, map variables in the data to visual properties of the geom (**aesthetics**) like **size**, **color**, and **x** and **y** locations.



Complete the template below to build a graph.

```
ggplot(data = <DATA>) +  
  <GEOM_FUNCTION>(mapping = aes(<MAPPINGS>),  
    stat = <STAT>, position = <POSITION>) +  
  <COORDINATE_FUNCTION> +  
  <FACET_FUNCTION> +  
  <SCALE_FUNCTION> +  
  <THEME_FUNCTION>
```

required
Not required, sensible defaults supplied

ggplot(data = mpg, aes(x = cty, y = hwy)) Begins a plot that you finish by adding layers to. Add one geom function per layer.

last_plot() Returns the last plot.

ggsave("plot.png", width = 5, height = 5) Saves last plot as 5" x 5" file named "plot.png" in working directory. Matches file type to file extension.

Aes

Common aesthetic values.

color and fill - string ("red", "#RRGGBB")

linetype - integer or string (0 = "blank", 1 = "solid", 2 = "dashed", 3 = "dotted", 4 = "dotted", 5 = "longdash", 6 = "twodash")

size - integer (line width in mm)

shape - integer/shape name or a single character ("a")



Geoms

Use a geom function to represent data points, use the geom's aesthetic properties to represent variables. Each function returns a layer.

GRAPHICAL PRIMITIVES

a <- ggplot(economics, aes(date, unemployment))
b <- ggplot(seals, aes(x = long, y = lat))

a + **geom_blank()** and a + **expand_limits()**
Ensure limits include values across all plots.
b + **geom_curve**(aes(yend = lat + 1, xend = long + 1), curvature = 1) - x, xend, y, yend, alpha, angle, color, curvature, linetype, size
a + **geom_path**(lineend = "butt", linejoin = "round", linemitre = 1) - x, y, alpha, color, group, linetype, size
a + **geom_polygon**(aes(alpha = 50)) - x, y, alpha, color, fill, group, subgroup, linetype, size
b + **geom_rect**(aes(xmin = long, ymin = lat, xmax = long + 1, ymax = lat + 1)) - xmax, xmin, ymax, ymin, alpha, color, fill, linetype, size
a + **geom_ribbon**(aes(ymin = unemployment - 900, ymax = unemployment + 900)) - x, ymax, ymin, alpha, color, fill, group, linetype, size

LINE SEGMENTS

common aesthetics: x, y, alpha, color, linetype, size

b + **geom_abline**(aes(intercept = 0, slope = 1))
b + **geom_hline**(aes(yintercept = lat))
b + **geom_vline**(aes(xintercept = long))
b + **geom_segment**(aes(yend = lat + 1, xend = long + 1))
b + **geom_spoke**(aes(angle = 1:1155, radius = 1))

ONE VARIABLE continuous

c <- ggplot(mpg, aes(hwy)); c2 <- ggplot(mpg)

c + **geom_area**(stat = "bin") - x, y, alpha, color, fill, linetype, size
c + **geom_density**(kernel = "gaussian") - x, y, alpha, color, fill, group, linetype, size, weight
c + **geom_dotplot**() - x, y, alpha, color, fill
c + **geom_freqpoly**() - x, y, alpha, color, group, linetype, size
c + **geom_histogram**(binwidth = 5) - x, y, alpha, color, fill, linetype, size, weight
c2 + **geom_qq**(aes(sample = hwy)) - x, y, alpha, color, fill, linetype, size, weight

discrete

d <- ggplot(mpg, aes(f))

d + **geom_bar**() - x, alpha, color, fill, linetype, size, weight

TWO VARIABLES both continuous

e <- ggplot(mpg, aes(cty, hwy))

e + **geom_label**(aes(label = cty), nudge_x = 1, nudge_y = 1) - x, y, label, alpha, angle, color, family, fontface, hjust, lineheight, size, vjust
e + **geom_point**() - x, y, alpha, color, fill, shape, size, stroke
e + **geom_quantile**() - x, y, alpha, color, group, linetype, size, weight
e + **geom_rug**(sides = "bl") - x, y, alpha, color, linetype, size
e + **geom_smooth**(method = "lm") - x, y, alpha, color, fill, group, linetype, size, weight
e + **geom_text**(aes(label = cty), nudge_x = 1, nudge_y = 1) - x, y, label, alpha, angle, color, family, fontface, hjust, lineheight, size, vjust

one discrete, one continuous

f <- ggplot(mpg, aes(class, hwy))

f + **geom_col**() - x, y, alpha, color, fill, group, linetype, size
f + **geom_boxplot**() - x, y, lower, middle, upper, ymax, ymin, alpha, color, fill, group, linetype, shape, size, weight
f + **geom_dotplot**(binaxis = "y", stackdir = "center") - x, y, alpha, color, fill, group
f + **geom_violin**(scale = "area") - x, y, alpha, color, fill, group, linetype, size, weight

both discrete

g <- ggplot(diamonds, aes(cut, color))

g + **geom_count**() - x, y, alpha, color, fill, shape, size, stroke
e + **geom_jitter**(height = 2, width = 2) - x, y, alpha, color, fill, shape, size

THREE VARIABLES

sealsSz <- with(seals, sqrt(delta_long^2 + delta_lat^2)); l <- ggplot(seals, aes(long, lat))

l + **geom_contour**(aes(z = sz)) - x, y, z, alpha, color, group, linetype, size, weight
l + **geom_contour_filled**(aes(fill = z)) - x, y, alpha, color, fill, group, linetype, size, subgroup

continuous bivariate distribution

h <- ggplot(diamonds, aes(carat, price))

h + **geom_bin2d**(binwidth = c(0.25, 500)) - x, y, alpha, color, fill, linetype, size, weight
h + **geom_density_2d**() - x, y, alpha, color, group, linetype, size
h + **geom_hex**() - x, y, alpha, color, fill, size

continuous function

i <- ggplot(economics, aes(date, unemployment))

i + **geom_area**() - x, y, alpha, color, fill, linetype, size
i + **geom_line**() - x, y, alpha, color, group, linetype, size
i + **geom_step**(direction = "hv") - x, y, alpha, color, group, linetype, size

visualizing error

df <- data.frame(grp = c("A", "B"), fit = 4:5, se = 1:2)
j <- ggplot(df, aes(grp, fit, ymin = fit - se, ymax = fit + se))

j + **geom_crossbar**(fatten = 2) - x, y, ymax, ymin, alpha, color, fill, group, linetype, size

j + **geom_errorbar**() - x, ymax, ymin, alpha, color, group, linetype, size, width
Also **geom_errorbarh**()

j + **geom_linerange**() - x, ymin, ymax, alpha, color, group, linetype, size

j + **geom_pointrange**() - x, y, ymin, ymax, alpha, color, fill, group, linetype, shape, size

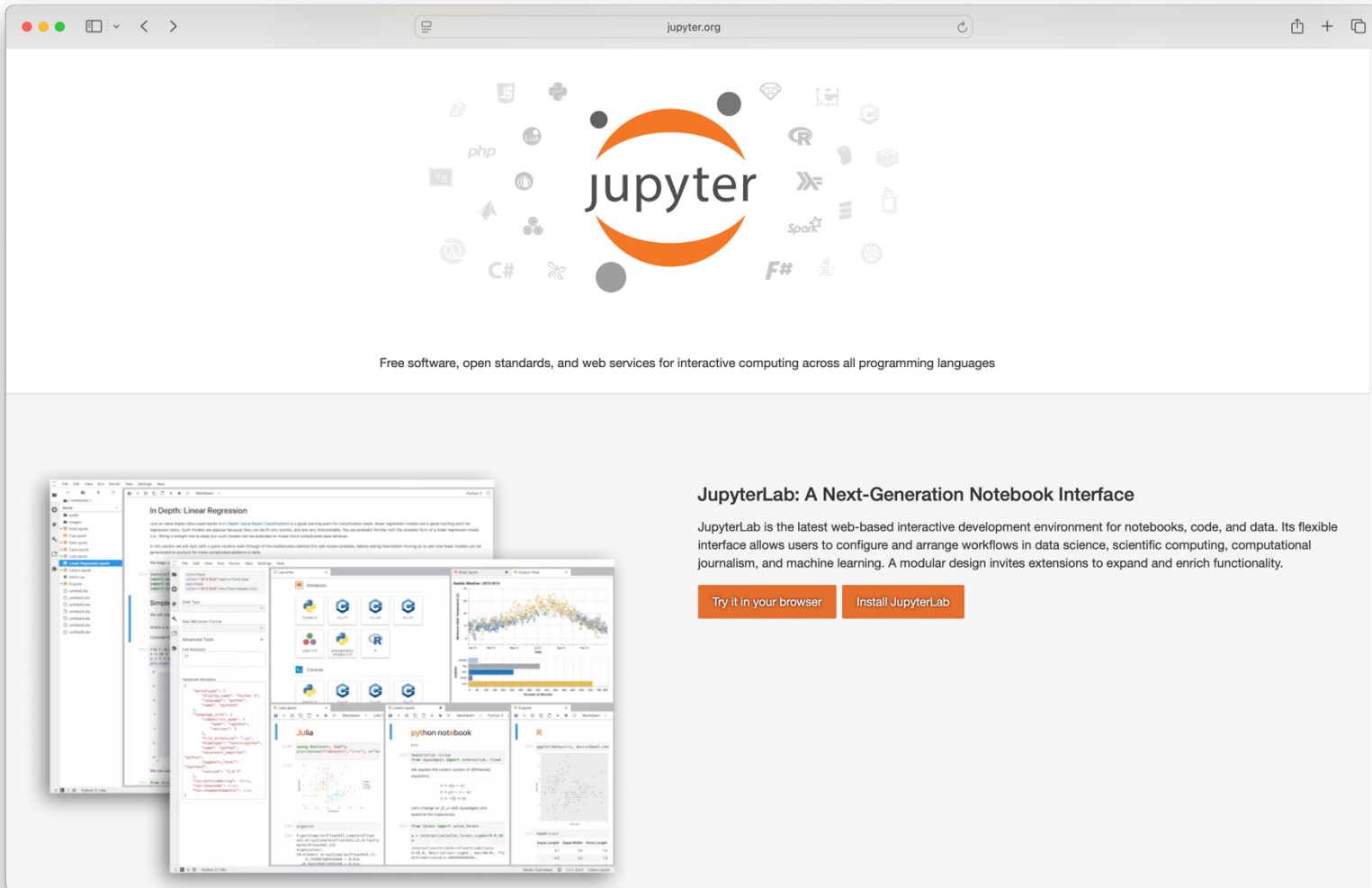
maps

data <- data.frame(murder = USArrests\$Murder, state = tolower(rownames(USArrests)))

map <- map_data("state")
k <- ggplot(data, aes(fill = murder))

k + **geom_map**(aes(map_id = state), map = map) + **expand_limits**(x = map\$long, y = map\$lat)
map_id, alpha, color, fill, linetype, size

Jupyter Notebooks



The image shows a browser window at jupyter.org. The main heading is "jupyter" with a large orange circular logo. Below it, the text reads: "Free software, open standards, and web services for interactive computing across all programming languages". At the bottom, there is a collage of various JupyterLab notebooks. One notebook titled "In Depth: Linear Regression" is visible, showing a scatter plot and a linear regression line. Other notebooks show code snippets in Python, Julia, and R, along with various data visualizations like bar charts and scatter plots.

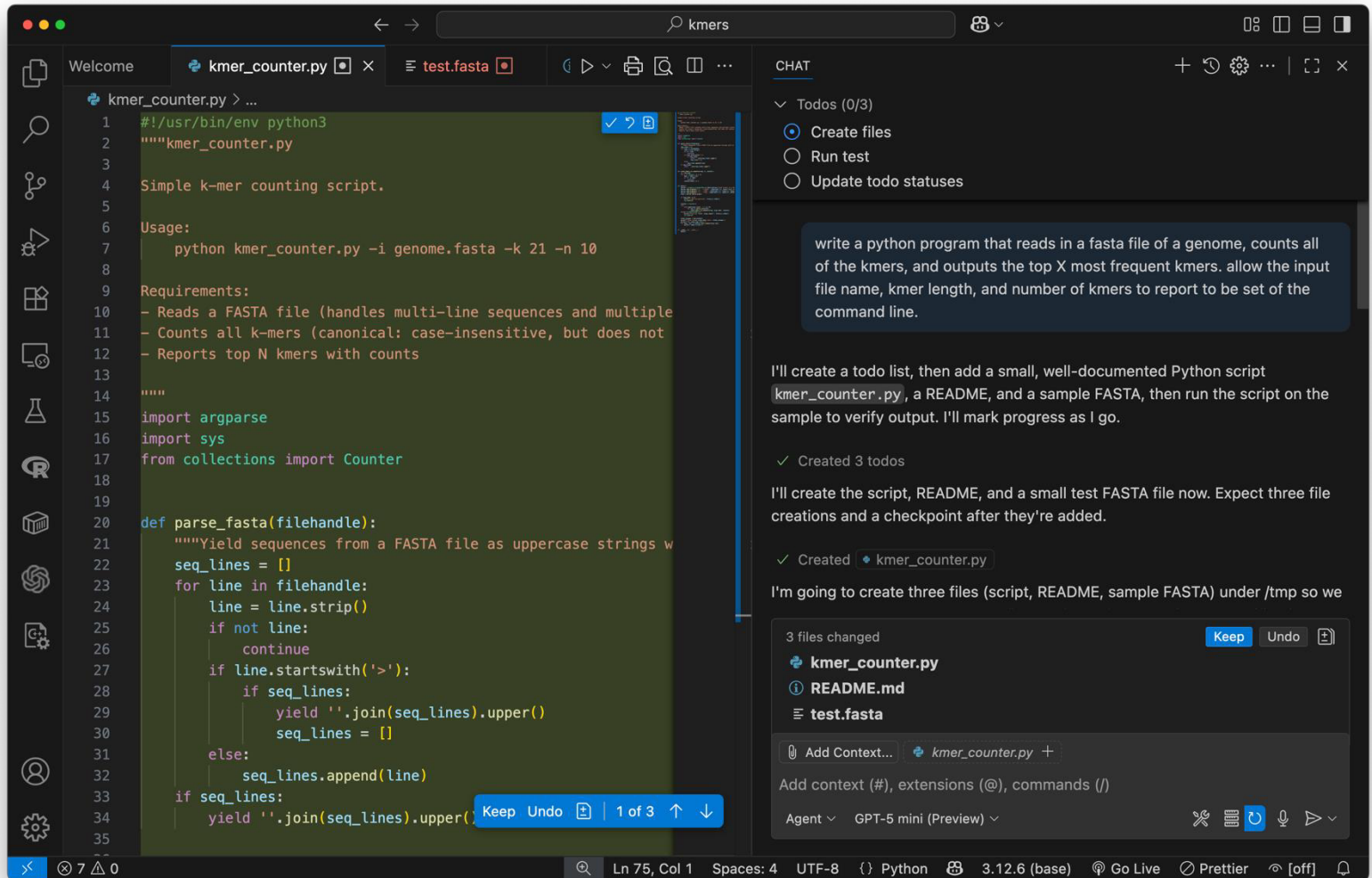
JupyterLab: A Next-Generation Notebook Interface

JupyterLab is the latest web-based interactive development environment for notebooks, code, and data. Its flexible interface allows users to configure and arrange workflows in data science, scientific computing, computational journalism, and machine learning. A modular design invites extensions to expand and enrich functionality.

[Try it in your browser](#) [Install JupyterLab](#)

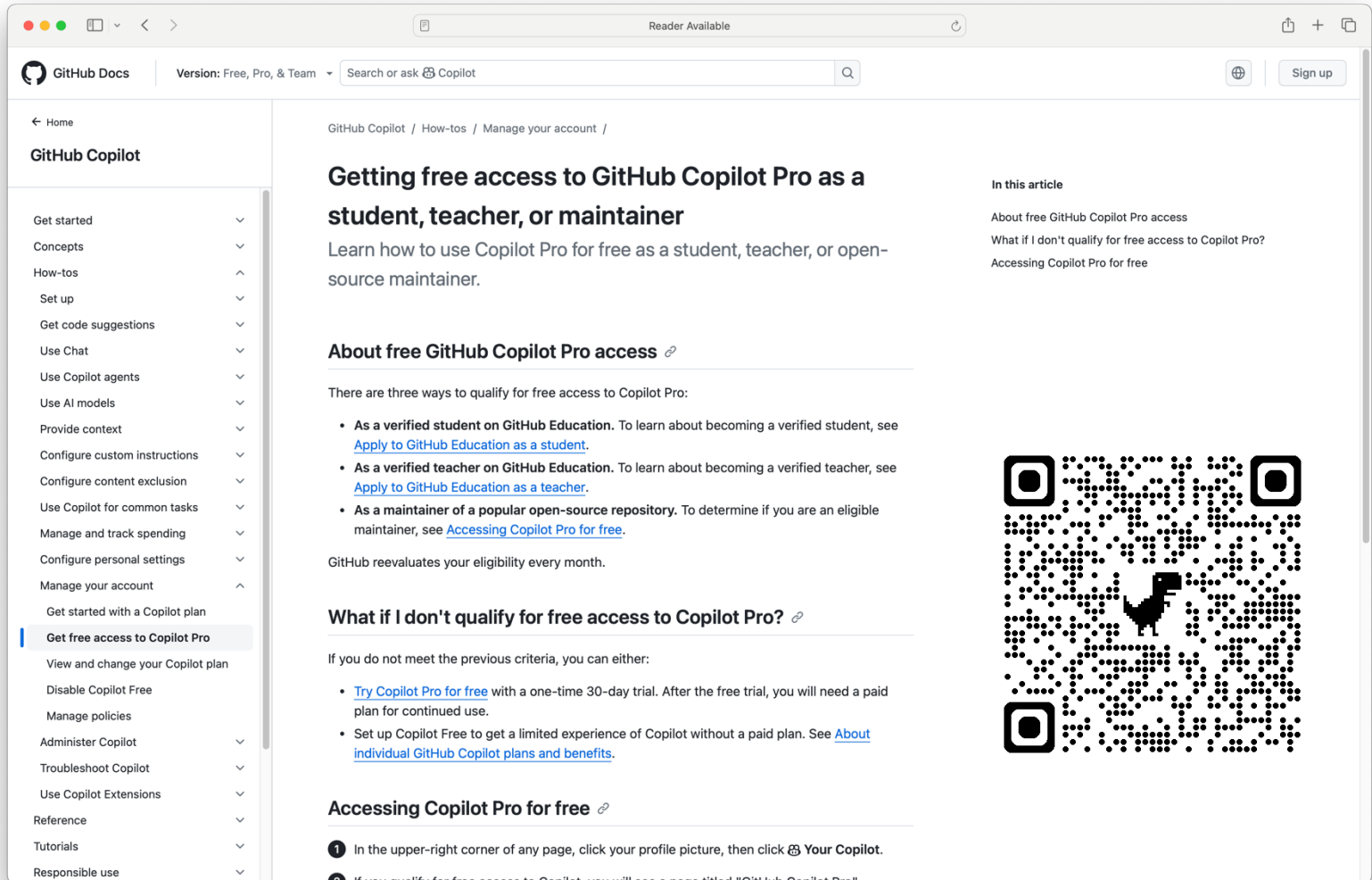
<https://jupyter.org>

VS Code



“write a python program that reads in a fasta file of a genome, counts all of the kmers, and outputs the top X most frequent kmers. allow the input file name, kmer length, and number of kmers to report to be set of the command line.”

Github Copilot Pro



The screenshot shows a web browser window displaying the GitHub Docs page for 'Getting free access to GitHub Copilot Pro as a student, teacher, or maintainer'. The page is titled 'Getting free access to GitHub Copilot Pro as a student, teacher, or maintainer' and includes a sub-header 'Learn how to use Copilot Pro for free as a student, teacher, or open-source maintainer.' The left sidebar shows the 'GitHub Copilot' section with a list of topics, including 'Get free access to Copilot Pro' which is highlighted. The main content area lists three ways to qualify for free access to Copilot Pro: as a verified student on GitHub Education, as a verified teacher on GitHub Education, and as a maintainer of a popular open-source repository. A QR code is displayed on the right side of the page. The browser's address bar shows the URL 'https://docs.github.com/en/copilot/how-tos/manage-your-account/get-free-access-to-copilot-pro'.

GitHub Docs | Version: Free, Pro, & Team | Search or ask Copilot | Sign up

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GitHub Copilot

- Get started
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- How-tos
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- Get code suggestions
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- Provide context
- Configure custom instructions
- Configure content exclusion
- Use Copilot for common tasks
- Manage and track spending
- Configure personal settings
- Manage your account
- Get started with a Copilot plan
- Get free access to Copilot Pro**
- View and change your Copilot plan
- Disable Copilot Free
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- Administer Copilot
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Getting free access to GitHub Copilot Pro as a student, teacher, or maintainer

Learn how to use Copilot Pro for free as a student, teacher, or open-source maintainer.

About free GitHub Copilot Pro access

There are three ways to qualify for free access to Copilot Pro:

- As a **verified student on GitHub Education**. To learn about becoming a verified student, see [Apply to GitHub Education as a student](#).
- As a **verified teacher on GitHub Education**. To learn about becoming a verified teacher, see [Apply to GitHub Education as a teacher](#).
- As a **maintainer of a popular open-source repository**. To determine if you are an eligible maintainer, see [Accessing Copilot Pro for free](#).

GitHub reevaluates your eligibility every month.

What if I don't qualify for free access to Copilot Pro?

If you do not meet the previous criteria, you can either:

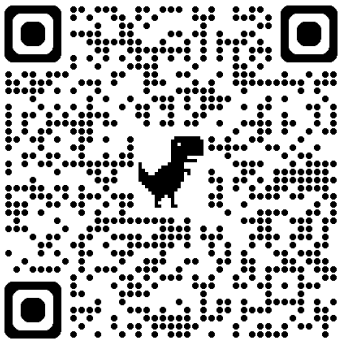
- [Try Copilot Pro for free](#) with a one-time 30-day trial. After the free trial, you will need a paid plan for continued use.
- Set up Copilot Free to get a limited experience of Copilot without a paid plan. See [About individual GitHub Copilot plans and benefits](#).

Accessing Copilot Pro for free

- In the upper-right corner of any page, click your profile picture, then click **Your Copilot**.
- If you qualify for free access to Copilot, you will see a page titled "GitHub Copilot Pro".

In this article

- About free GitHub Copilot Pro access
- What if I don't qualify for free access to Copilot Pro?
- Accessing Copilot Pro for free



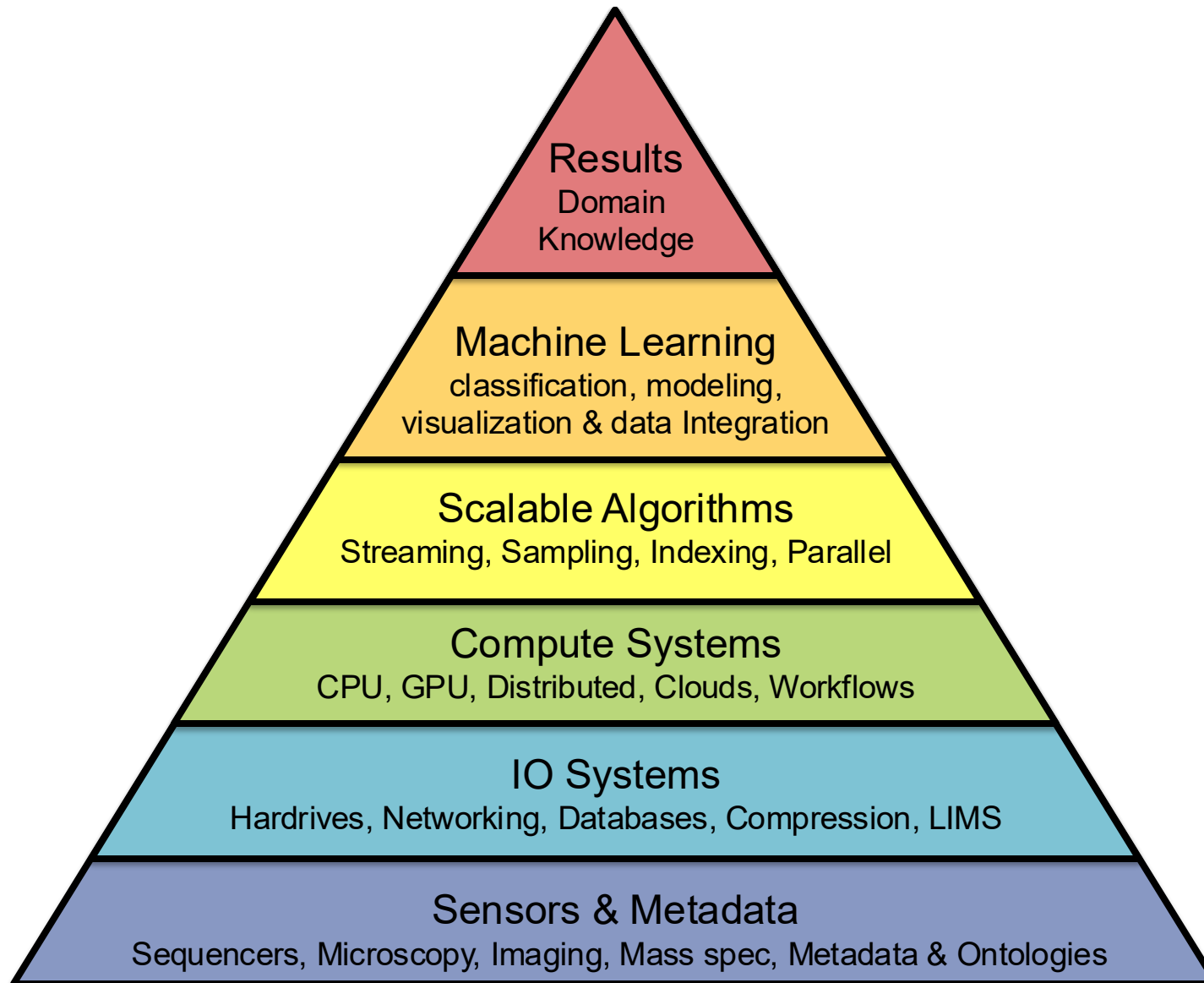
<https://docs.github.com/en/copilot/how-tos/manage-your-account/get-free-access-to-copilot-pro>

Unsolved Questions in Biology

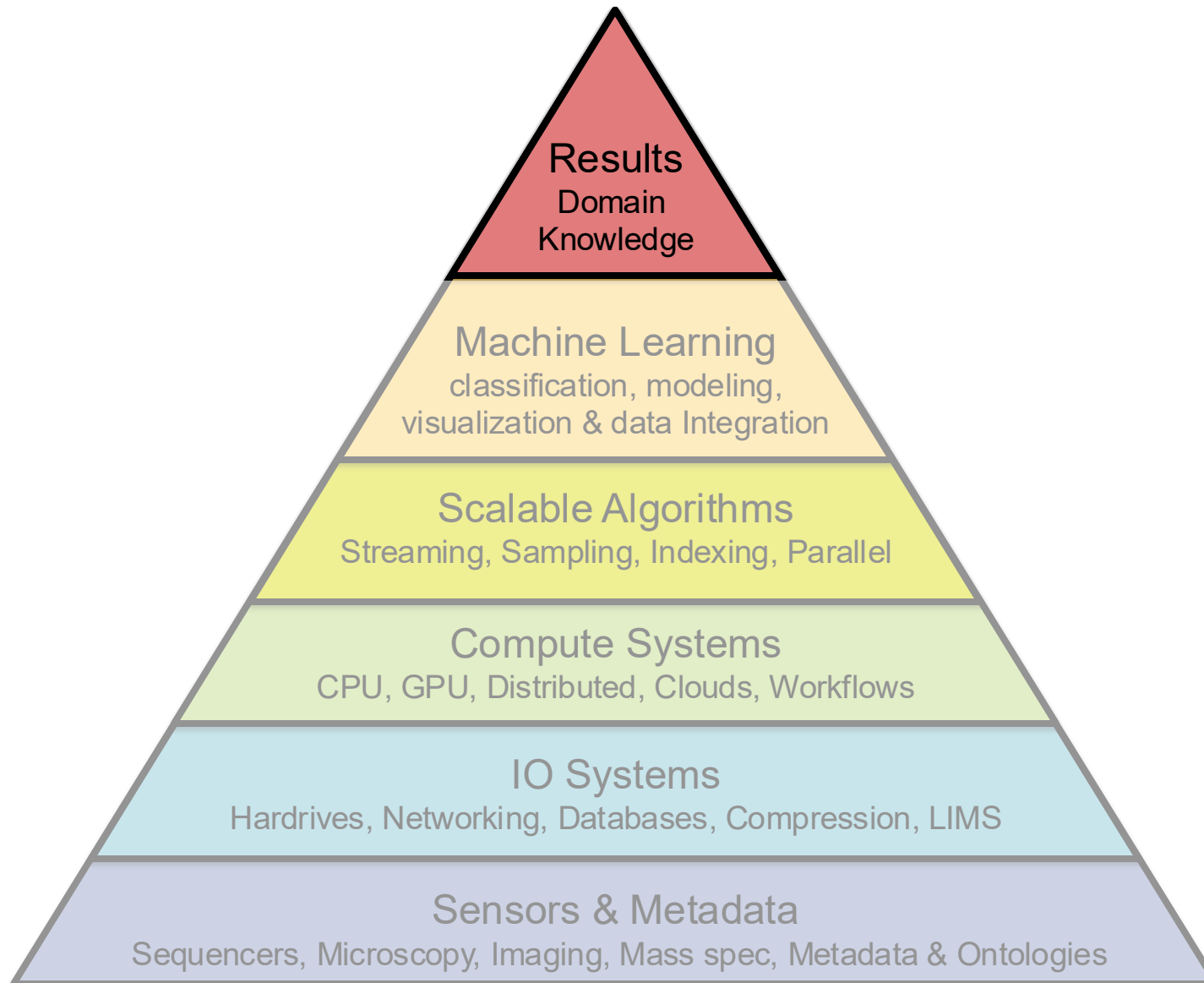
- What is your genome sequence?
- How does your genome compare to my genome?
- Where are the genes and how active are they?
- How does gene activity change during development?
- How does splicing change during development?
- How does methylation change during development?
- How does chromatin change during development?
- How is your genome folded in the cell?
- Where do proteins bind and regulate genes?
- What virus and microbes are living inside you?
- How do your mutations relate to disease?
- What drugs and treatments should we give you?
- ***Plus thousands and thousands more***



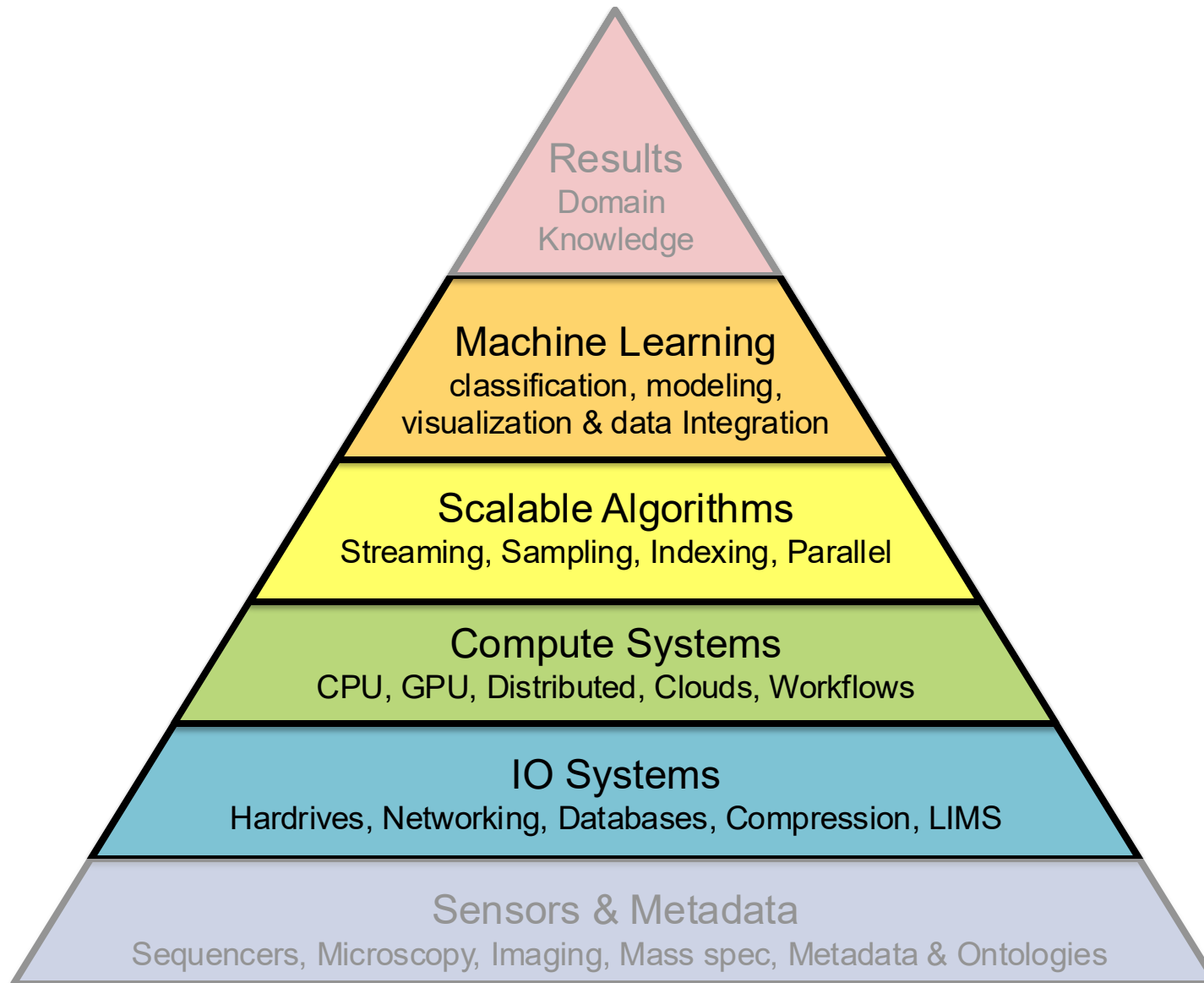
Comparative Genomics Technologies



Comparative Genomics Technologies



Comparative Genomics Technologies

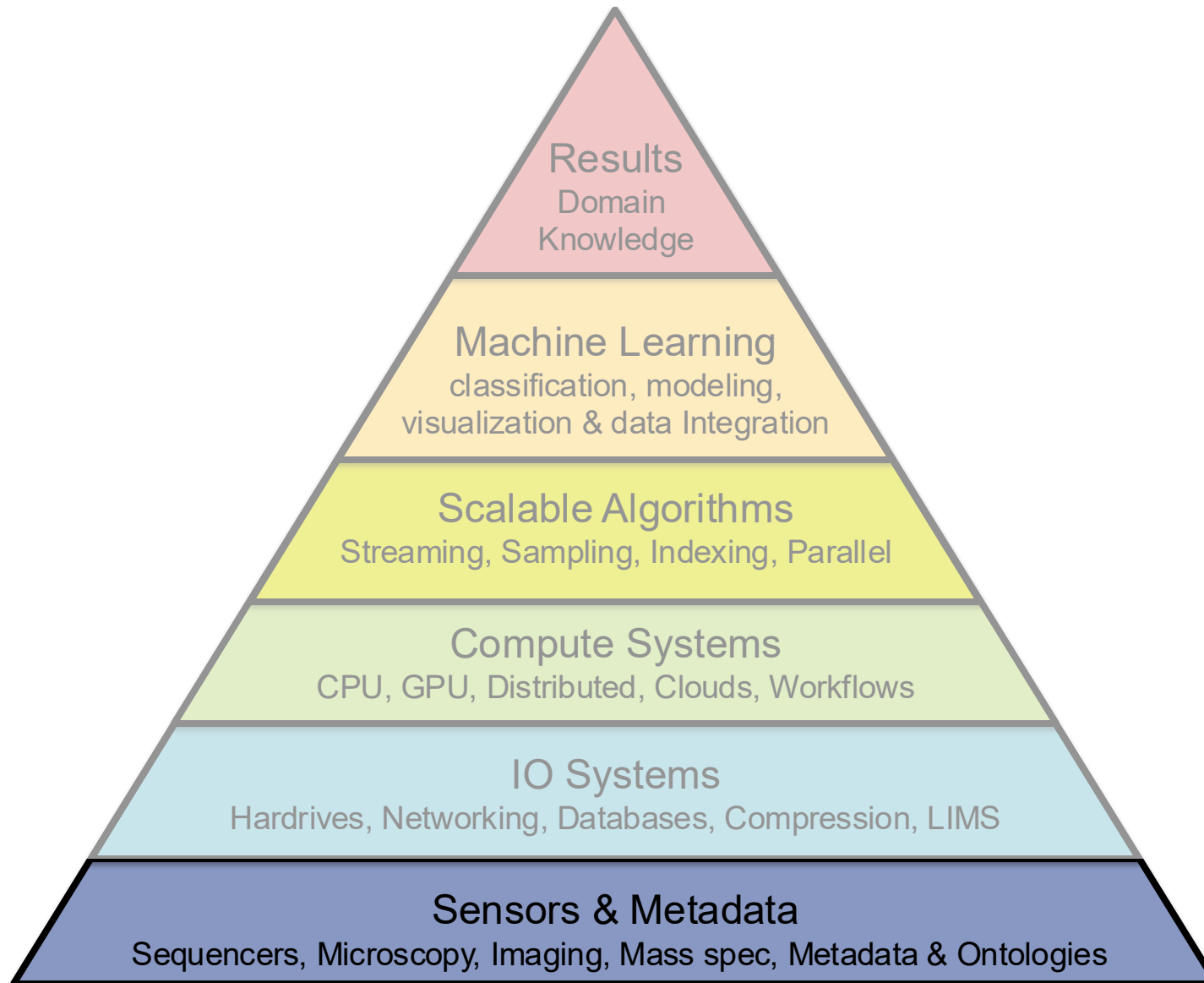


Selected Topics

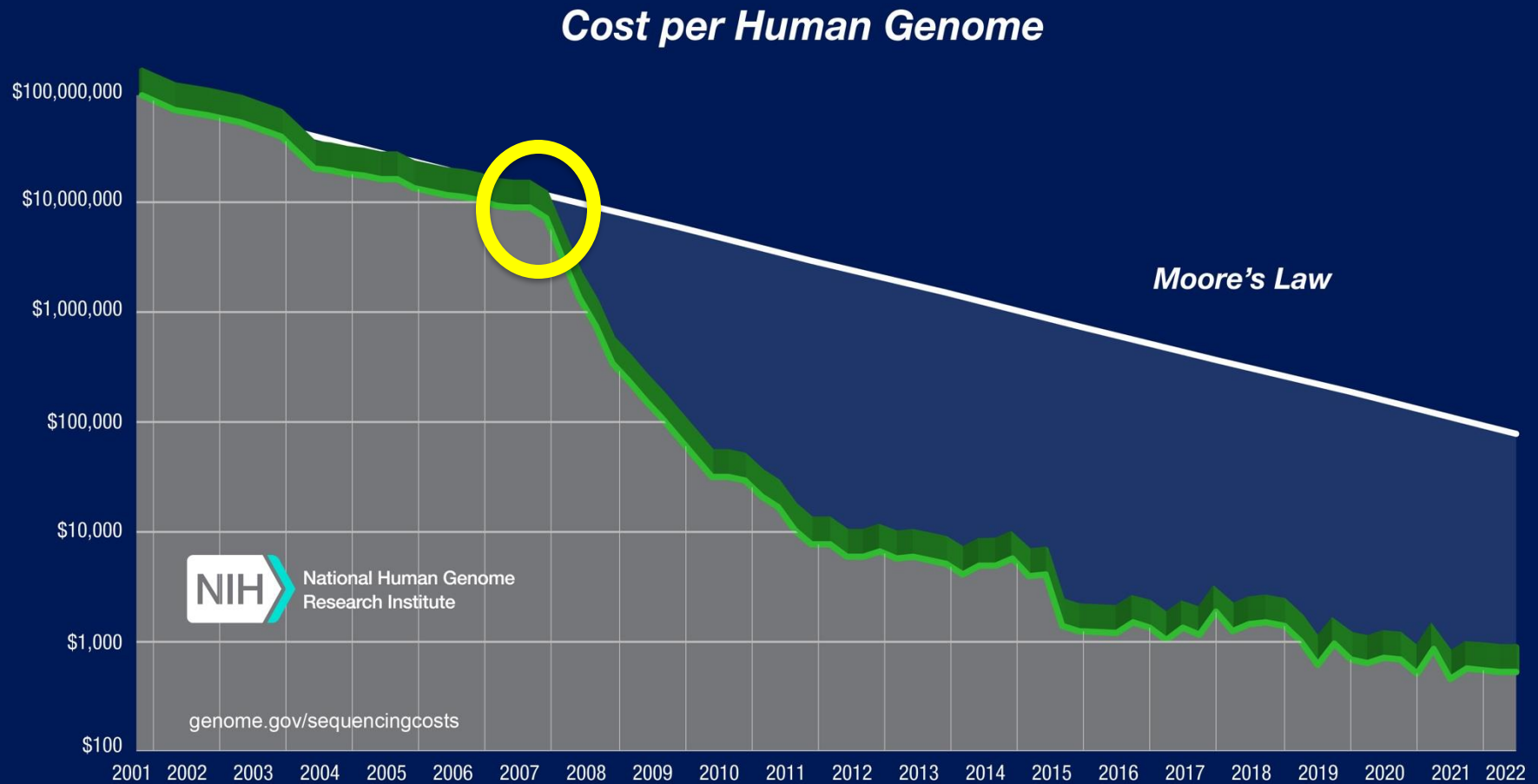
- Genome assembly, whole genome alignment
- Full text indexing: Suffix Trees, Suffix Arrays, FM-index
- Dynamic Programming: Edit Distance, sequence similarity
- Read mapping & Variant identification
- Gene Finding: HMMs, Plane-sweep algorithms
- RNA-seq: mapping, assembly, quantification
- ChIP-seq: Peak finding, motif finding
- Methylation-seq: Mapping, CpG island detection
- HiC: Domain identification, scaffolding
- Chromatin state analysis: ChromHMM, Enformer, etc
- Scalable genomics: Cloud computing, scalable data structures
- Population & single cell analysis: clustering, pseudotime
- Disease analysis, cancer genomics, Metagenomics
- Deep learning in genomics



Comparative Genomics Technologies



Cost per Genome

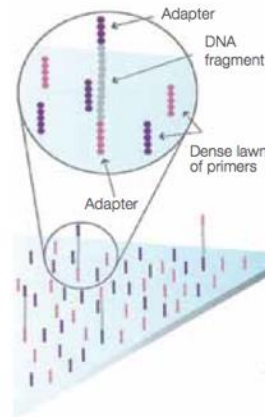


Second Generation Sequencing

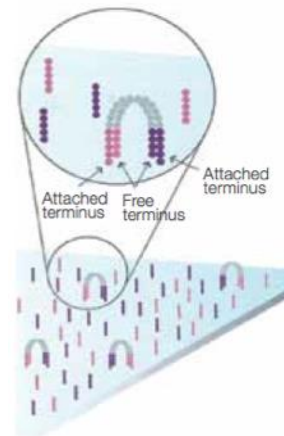


Illumina NovaSeq 6000
Sequencing by Synthesis

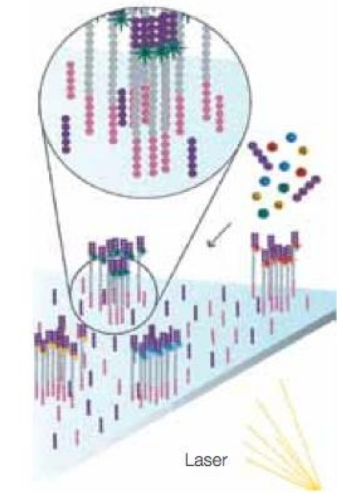
>3Tbp / day
(JHU has 4 of these!)



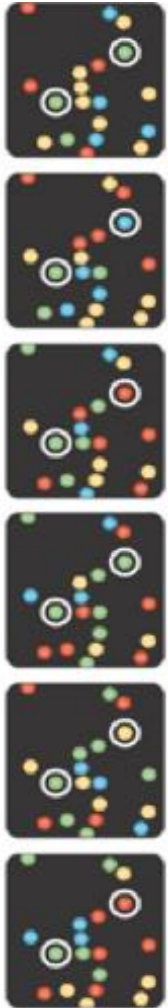
1. Attach



2. Amplify

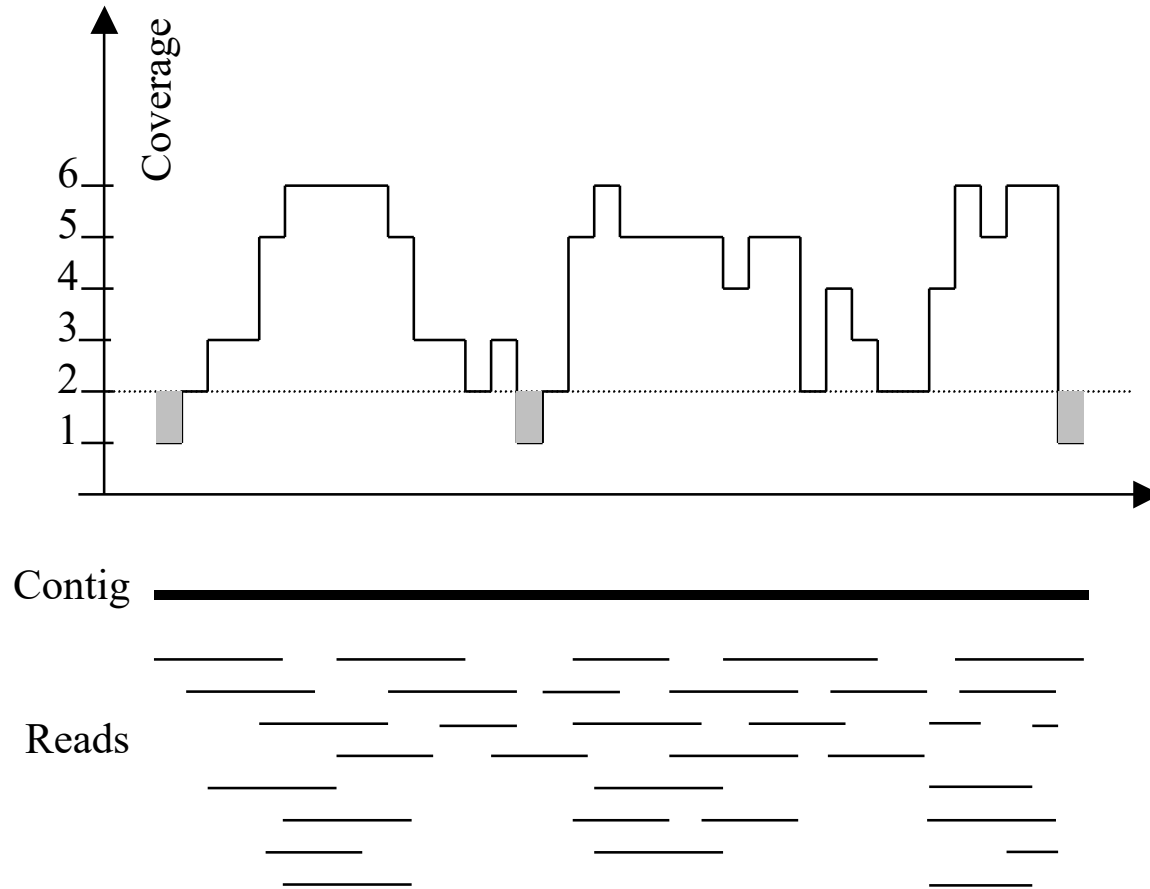


3. Image



Metzker (2010) Nature Reviews Genetics 11:31-46
<https://www.youtube.com/watch?v=fCd6B5HRaZ8>

Typical sequencing coverage

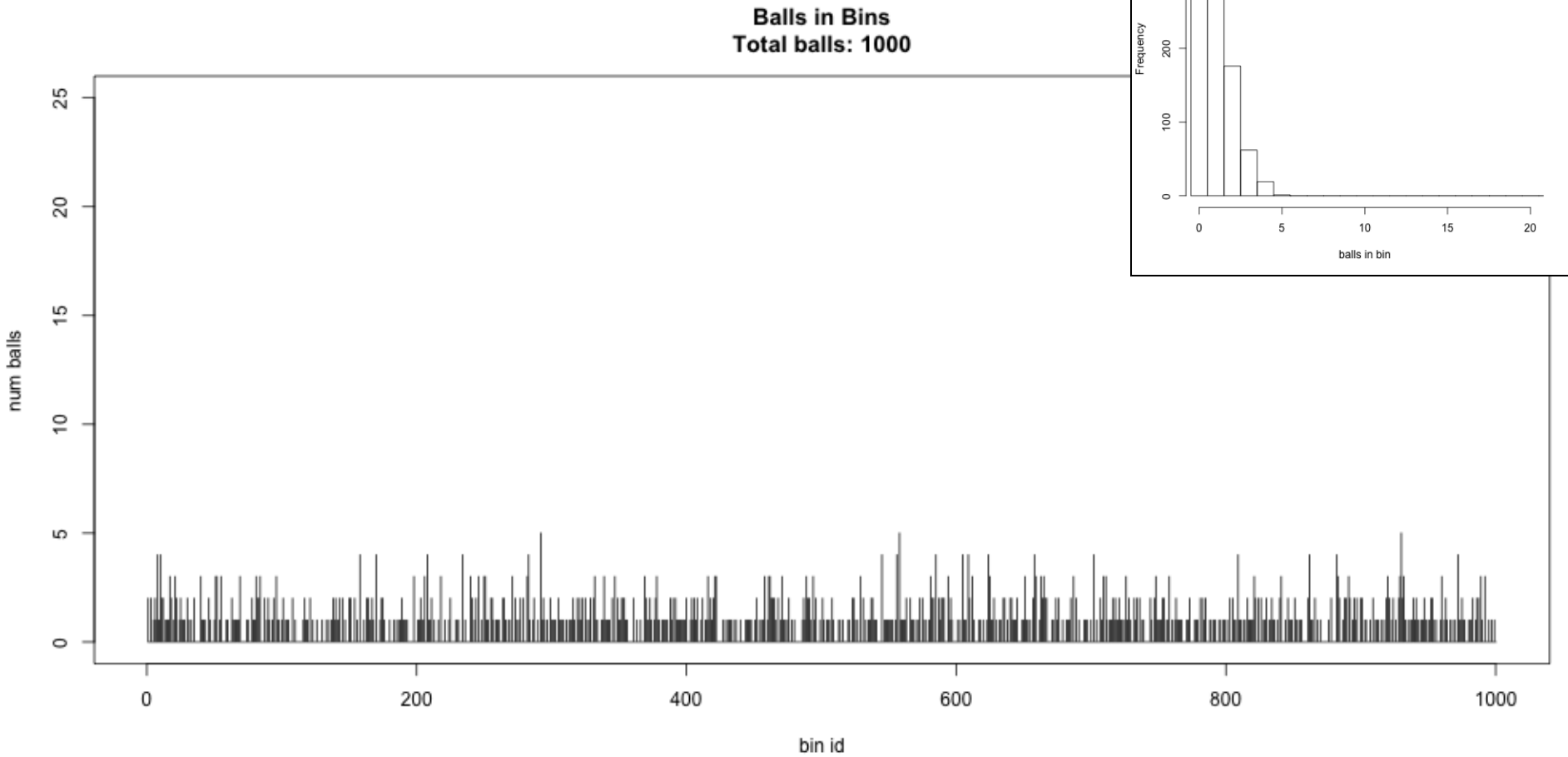


Imagine raindrops on a sidewalk

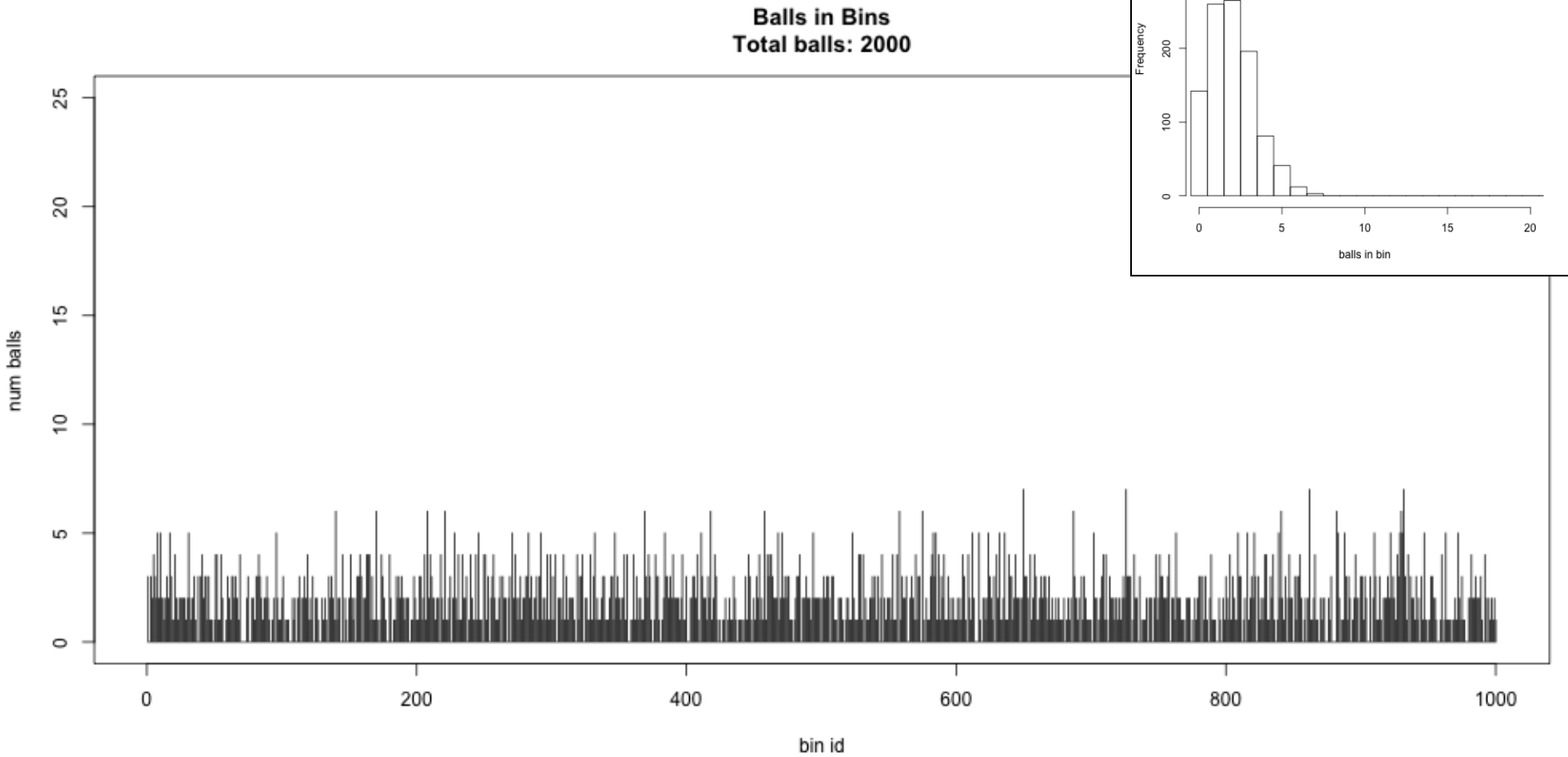
We want to cover the entire sidewalk but each drop costs \$1

If the genome is 10 Mbp, should we sequence 100k 100bp reads?

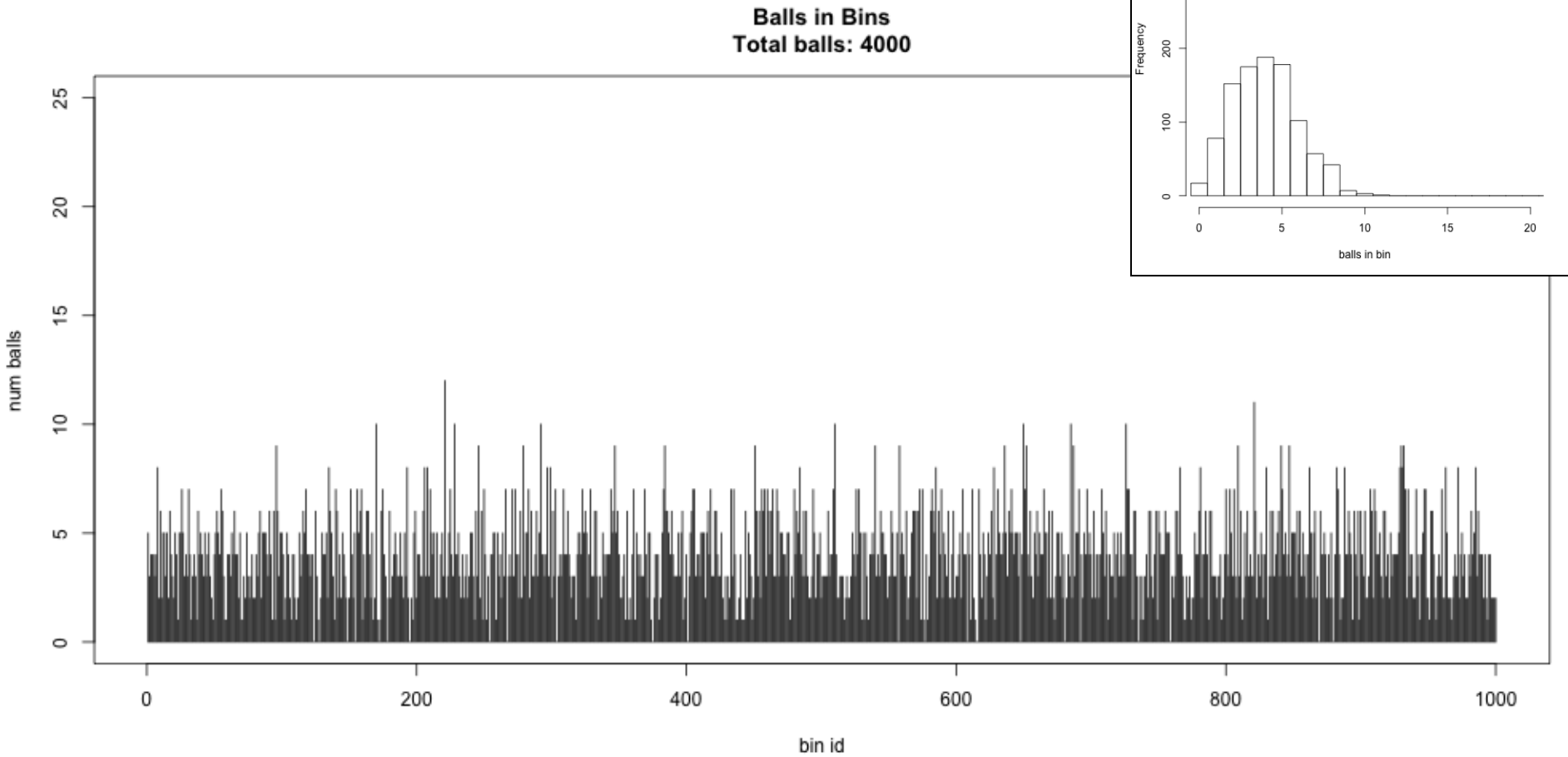
Ix sequencing



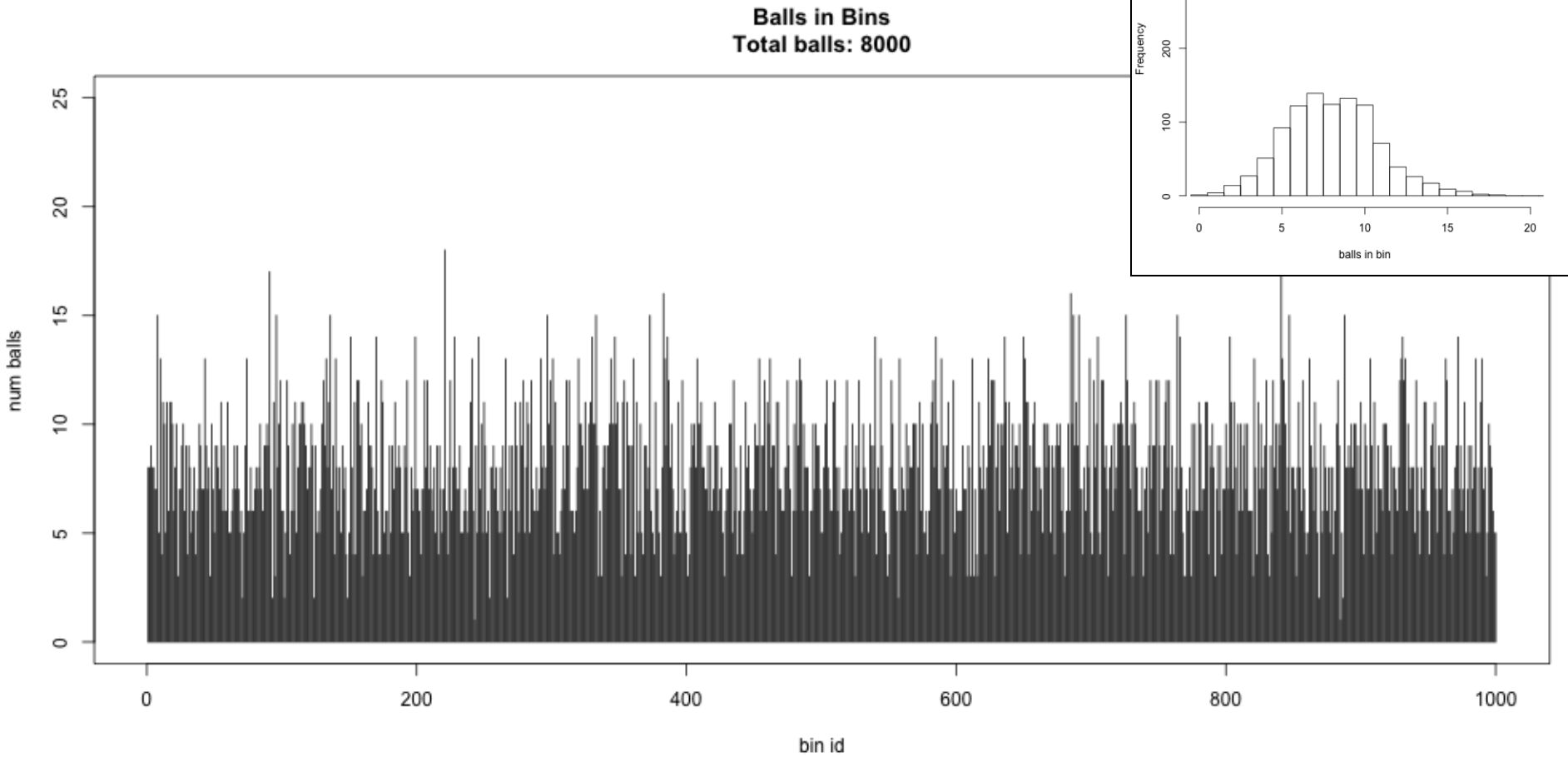
2x sequencing



4x sequencing



8x sequencing



Poisson Distribution

The probability of a given number of events occurring in a fixed interval of time and/or space if these events occur with a known average rate and independently of the time since the last event.

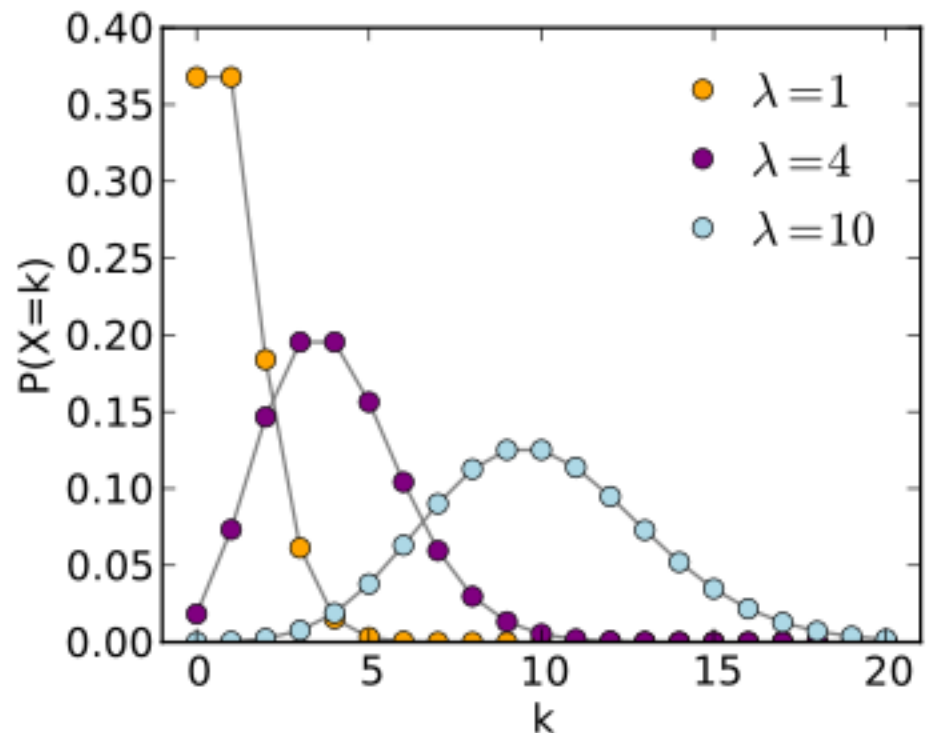
Formulation comes from the limit of the binomial equation

Resembles a normal distribution, but over the positive values, and with only a single parameter.

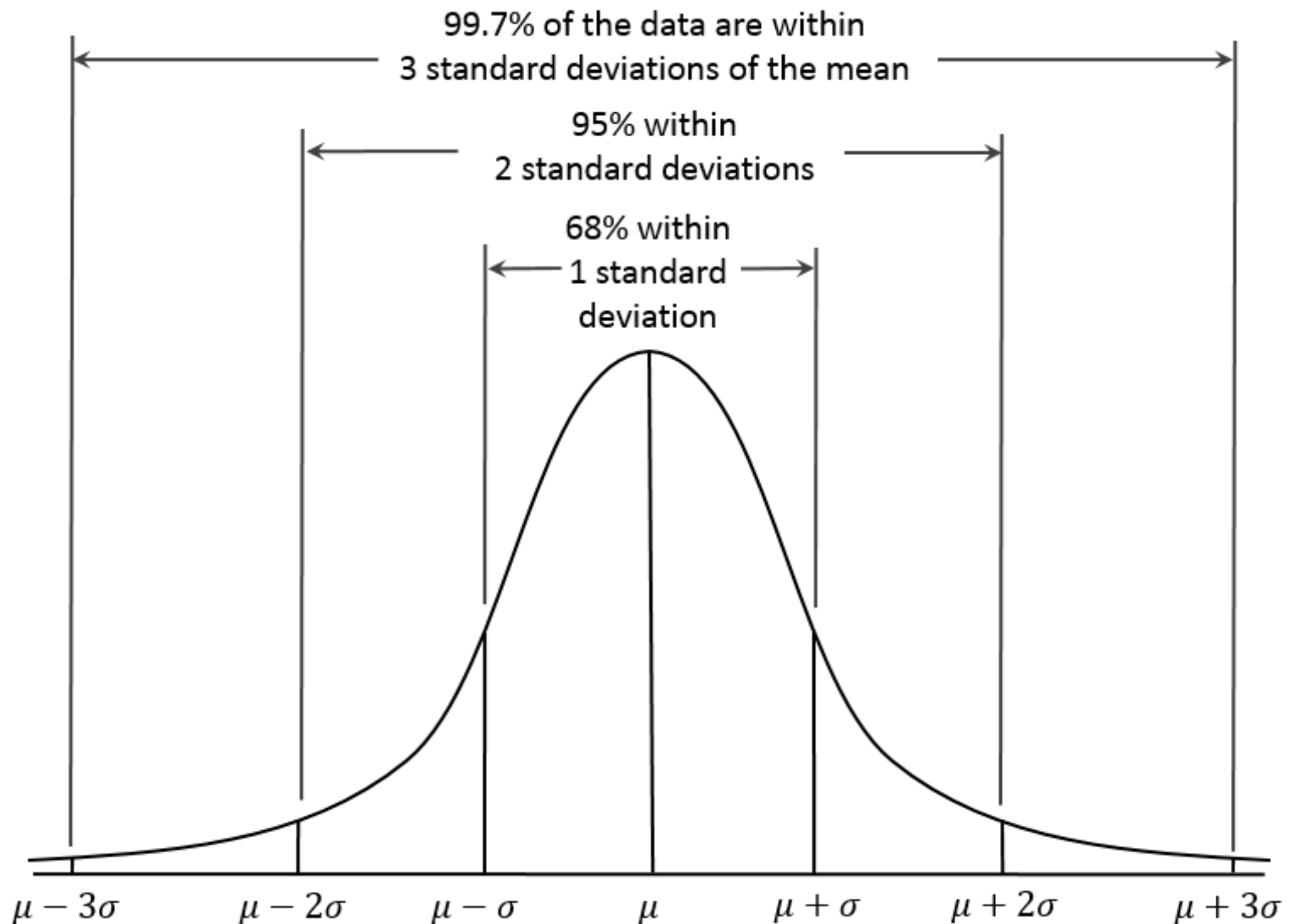
Key properties:

- ***The standard deviation is the square root of the mean.***
- ***For mean > 5, well approximated by a normal distribution***

$$P(k) = \frac{\lambda^k}{k!} e^{-\lambda}$$



Normal Approximation



Can estimate Poisson distribution as a normal distribution when $\lambda > 10$